

HB - Salma (branch hierarchical model) — implemented equations (B1–B5)

The independent "hidden-confidence" observer from the branch `new_implementation_with_hierarchical_fitting` (`independent_transcript_paper_model/src/hierarchical_confidence/model.py`). Class docstring: "Discrete hidden-confidence observer with one integrated posterior." Labelled B1–B5 to line up with `hb_integration`'s M1–M5 and the switch's H1–H5, so all three documents read block-for-block.

Numerical grid (fixed support, NOT fitted). Unlike our $1^\circ / 360$ -point grid:

```
θ grid: 72 bins, linspace(0, 360, endpoint=False) = 0, 5, 10, ..., 355°
κ grid: [0.0] ∪ geomspace(0.05, 50, 15) = 16 support points
        (note: κ = 0 IS in the support → a uniform-prior component is
         reachable through the belief, with no explicit mixture weight)
```

Everything is computed in LOG space with `logsumexp` for stability. Building block — von Mises pmf on the grid, written stably as $\exp(\kappa(\cos(\theta - \mu) - 1))$ then row-normalised: $V(\theta; \mu, \kappa) \propto \exp(\kappa \cdot (\cos(\theta - \mu) - 1))$.

B1 — Prior basis (pure von Mises per κ — NO mixture floor)

For every κ on the grid, a normalised von Mises at the prior mean 225° :

```
prior_basis[κ, θ] = V(θ; 225, κ)
```

There is no $\alpha \cdot \text{VM} + (1-\alpha)/360$ mixture here. The only route to a "flat" prior is the $\kappa = 0$ support point (which makes V uniform). Source: `_prior_basis`.

B2 — Hidden-confidence trajectory over κ (geometric forget + feedback Bayes)

A belief $H_t(\kappa)$ over the κ grid, carried across ALL trials (never reset at block boundaries — an explicit pilot choice). Initialised uniform, $H_0(\kappa) = 1/n_\kappa$. Each trial, in log space:

```
predict (forget):  log H~_t = ρ · log H_{t-1}          (then renormalise)
record H~_t as the PRE-feedback state used for the estimate
correct (Bayes):   log H_t = log H~_t + logV(θ_fb; 225, κ)  (then renormalise)
```

- ρ (rho), $0 < \rho < 1$: geometric forgetting. Raising the log-belief to power ρ and renormalising shrinks it toward uniform — the smaller ρ , the faster the leak toward "no confidence". (Contrast: our M5 forgets LINEARLY toward `b_0`.)
- The feedback term is the von Mises log-likelihood of the shown direction θ_{fb} under prior mean 225° at each κ . Source: `confidence_trajectory(), _make_feedback_log_likelihood()`.

B3 — Effective prior = INTEGRATE the belief into ONE prior (THE key difference)

Marginalise κ BEFORE the read-out: collapse the per- κ prior basis over the current pre-feedback belief into a single effective prior pmf for the trial:

```
effective_prior_t(θ) = Σ_κ H~_t(κ) · V(θ; 225, κ)      (then renormalise)
```

This is the "one integrated posterior" of the docstring, and it is the exact axis on which this model differs from ours: `hb_integration` marginalises κ AFTER the read-out (average of per- κ read-outs, M4); this model marginalises κ BEFORE (one prior → one posterior → one read-out). Source: `effective_priors()`.

B4 — Posterior + tie-aware MAP read-out (one integrated posterior)

With sensory concentration `k_sensory(c)` (one per coherence), form the log posterior as sensory likelihood $\times$ the single effective prior, then take the MAP — but ties are handled explicitly (every tied maximum bin shares the read-out mass equally, rather than an arbitrary argmax):

```
log posterior_t(θ | m) = logP(θ | m, k_sensory(c)) + log effective_prior_t(θ)
```

```
MAP read-out: tied_map = { bins achieving the posterior max }, each weight 1/(#ties)
```

The coarse 72-bin grid makes exact MAP ties common, which is why the tie-aware treatment matters here (it is a genuine numerical difference from our 360-bin argmax). Source: `tie_aware_map_support()`, used in `predict_response_pmfs()` / `negative_log_likelihood()`.

B5 — Percept push-forward + motor noise + lapse

Marginalise the MAP read-out over the internal measurement distribution, convolve with von Mises motor noise (via FFT), and mix in the lapse floor:

$$\text{percept}_t(\text{est} \mid \theta_{\text{true}}) = \sum_m P(m \mid \theta_{\text{true}}, k_{\text{sensory}}) \cdot \text{MAP}(\text{est} \mid m)$$
$$p_t(\text{est}) = (1 - \text{lapse}) \cdot [\text{percept} \blacksquare V(\cdot \ ; \ 0, k_{\text{motor}})] + \text{lapse} \cdot (1/n_{\theta})$$

Source: `predict_response_pmfs()` (FFT convolution) and the lapse line.

Fitted parameters (6) — leaner than ours

<code>k_sensory(0.06), k_sensory(0.12), k_sensory(0.24)</code>	sensory reliability, per coherence
<code>p</code>	geometric forgetting / volatility
<code>k_motor</code>	motor-noise concentration
<code>lapse</code>	lapse rate

(No α , no per-block prior-width parameter — 6 total vs our 7.) As in ours, the prior concentration κ is NOT fitted: it is the learned latent, carried as the belief $H_t(\kappa)$ and updated every trial by B2.

How this contrasts with `hb_integration (M1–M5)`

Same hierarchical idea — learn a belief over the prior concentration κ from feedback, trial by trial — but three real definitional differences:

- Combination rule (B3 vs M4). This model INTEGRATES the belief into one prior before the read-out ($\sum_{\kappa} H(\kappa) \cdot V$, then one posterior, one MAP). Ours AVERAGES per- κ read-outs after the fact ($\sum_{\kappa} b(\kappa) \cdot R_{\kappa}$). These are different distributions in general; both still produce Girshick bimodality, but the far-from-prior shape differs (this is the divergence the block-phase comparison figure shows, mean TV ≈ 0.16 in the far band).
- Prior parameterisation. This model has NO mixture weight α : its prior is a pure von Mises whose width is entirely emergent from the learned belief (a uniform component is reachable only via the $\kappa = 0$ grid point). Ours keeps α as a fitted mixture weight on top of the learned κ . $\rightarrow$ 6 params vs 7.
- Forgetting law. This model forgets GEOMETRICALLY ($\log H \leftarrow p \cdot \log H$, shrink toward uniform). Ours forgets LINEARLY toward the initial belief ($b \leftarrow (1-\lambda)b + \lambda \cdot b_0$).

Engineering (not model definition): 72-bin grid vs our 360-bin; full log-space with `logsumexp`; tie-aware MAP vs plain `argmax`; Powell optimiser with a time budget and Latin-hypercube multi-start — a deliberately multistart/GPU-friendly rewrite.